Class Test 1

Electricity and Electronics EBN 122

7 August 2019

Test Information			
Maximum marks:	15	Full marks:	15
Duration of test:	30 min + (5 min for OCR)	Open/Closed book:	Closed
Total number of pages (this page included)		10	

Important

1. The examination regulations of the University of Pretoria apply.
2. Answer all questions on the OCR forms provided. The question number **01** in this question paper refers to the numeric field 01 of the numeric based OCR form.
3. Read the instructions on the OCR form carefully.
4. Where applicable, answers must include units.
5. Final answers must be written as real numbers and not as fractions.
6. Use at least 3 significant numbers to write irrational numbers.

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INSTRUCTIONS

Do step 0 in the first 5 minutes of the session. (-5 to 0 min)

Do step 1 in the first 30 minutes of the test. (0-30 min)

Do step 2 within the following 5 minutes. (30-35 min)

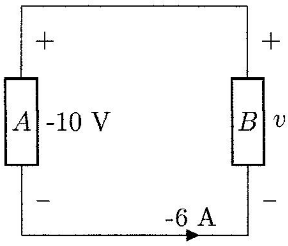
STEP 0 (-5 min to 0 min): Complete the front page of the OCR form.

STEP 1 (0 min to 30 min): Do your own calculations on the question paper. Write down the answers for numeric questions on the PRACTICE ANSWER SHEET. Be sure to include the units where applicable.

STEP 2 (30 min to 35 min): (only if numeric sheets are completed) Carefully copy the answers from the PRACTICE ANSWER SHEET to the OCR form.

- The Assessment ID is: **2019EBN122C01**
- Use a mechanical pencil with 0.5 mm HB lead to complete the form.
- Each cell should only have one character, and characters may not touch or cross the cell.
- **Do not use commas!** Please make use of full stops as decimal points.

Section 1

Questions 1 to 3 [3]	
Consider the circuit below and make use of the passive sign convention.	
	
01	Calculate the power P_B absorbed by the circuit element B. [1]
02	Is element A in the circuit supplying power (write a "1") or absorbing power (write a "2")? [1]
03	The circuit is used for 40 minutes. Calculate the energy E_A of the element A in kilowatt-hours. [1]

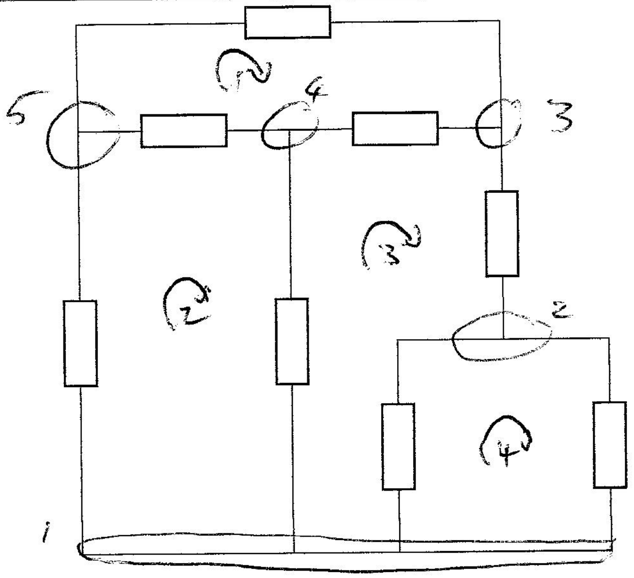
1. $P_B = -vi = -(-10)(-6) = -60\,\text{W}$

2. Power supplied = power absorbed

RHS element is supplying power (negative power), therefore LHS must be absorbing power. Answer: 2. Absorbing power.

3. $E_A = 60 \times \frac{40}{60} = 40\,\text{Wh} = 0.04\,\text{kWh}$

Question 4 [1]



04	Determine the number of independent loops l in the circuit above. [1]

Branches : 8

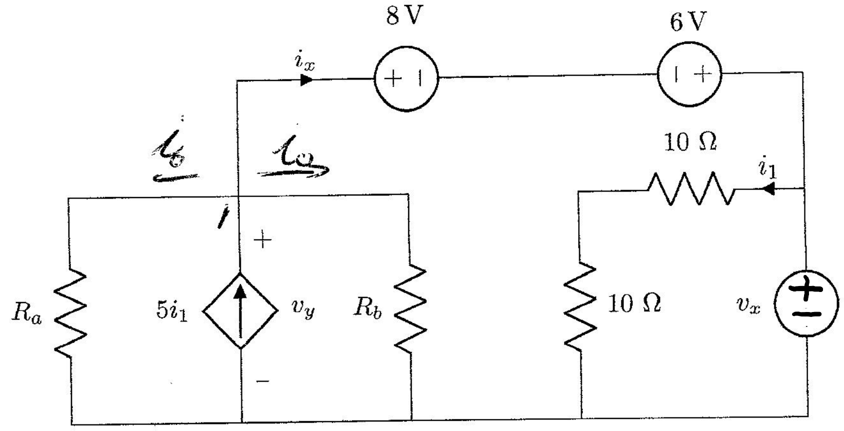
Nodes : 5

$b = l + n - 1$

$l = b - n + 1$

$= 8 - 5 + 1$

$= 4$ ⇒ Corresponds to observation

Questions 5 to 7 [3]	
Consider the circuit diagram below.	
	
05	Let $i_1 = 0.5$ A. Determine v_x. [1]
06	Determine v_y. [1]
07	Now assume that $i_1 = 1$ A, $v_y = 22$ V, $R_a = 10\ \Omega$, and $R_b = 10\ \Omega$. Use Kirchoff's current law (KCL) to determine i_x. [1]

5. $v_x = i_1(10+10) = 0.5 \times 20 = \underline{10\ V}$

6. $-v_y + 8 - 6 + v_x = 0$ (KVL round outer loop)

$-v_y + 2 + 10 = 0$

$v_y = \underline{12\ V}$

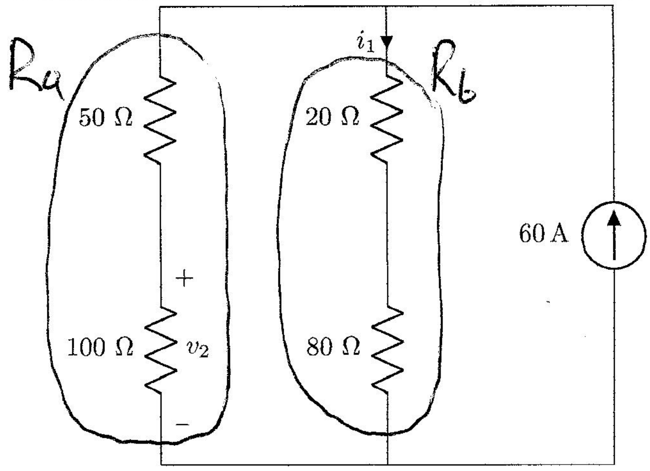
7. $i_x + 2i_o = 5i_1$ (KCL at node 1)

$i_x + 2\left(\frac{22}{10}\right) = 5$

$i_x + \frac{44}{10} = 5$

$i_x + 4.4 = 5$

$i_x = \underline{0.6\ A}$

Questions 8 to 9 [2]	
Consider the circuit diagram below.	
	
08	Determine i_1. [1]
09	Determine v_2. [1]

8. $R_a = 150\,\Omega$ $R_b = 100\,\Omega$

Current division: $i_1 = \frac{R_a}{R_a + R_b} 60$

$= 60\left(\frac{150}{250}\right)$

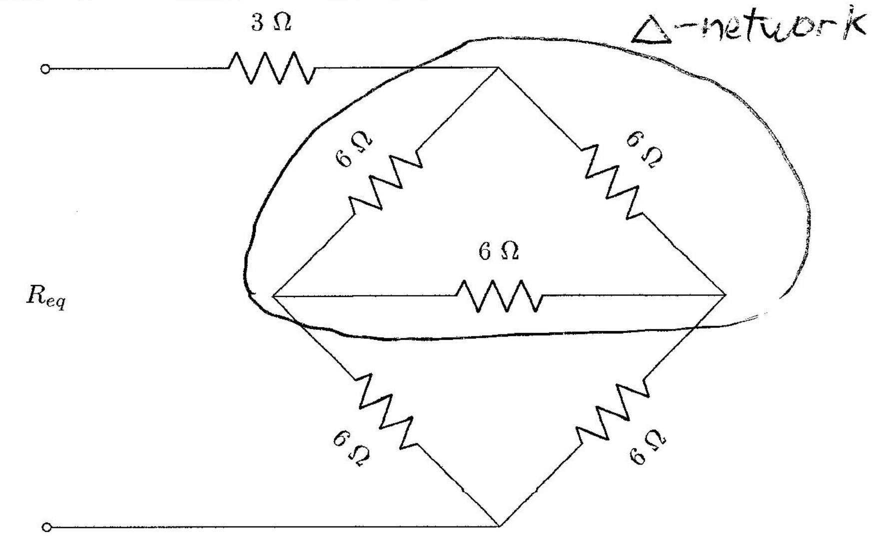
$= 36 \text{ A}$

$i_2 = 60 - 36 = 24 \text{ A}$ (From KCL, top node)

Ohm's law: $v_2 = 100 \times 24$

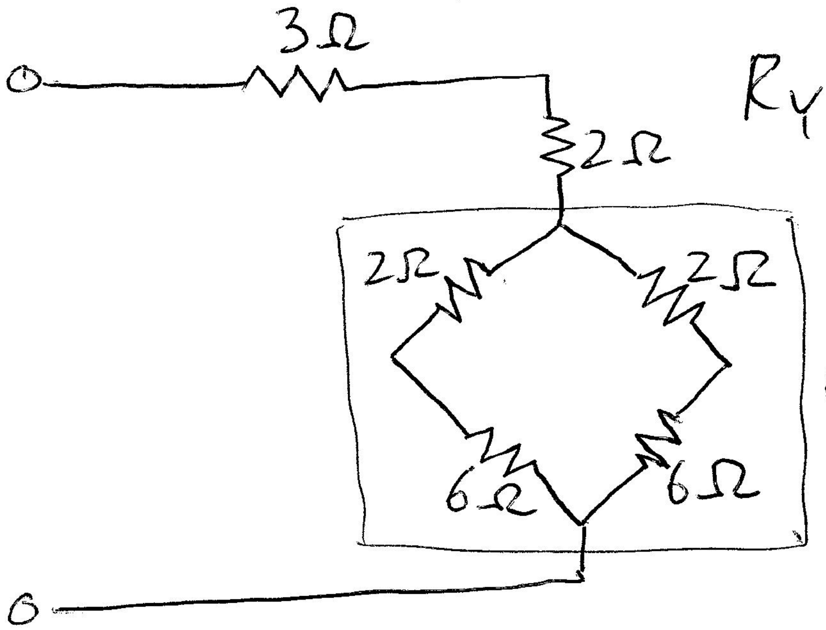
$= 2400 \text{ V}$

Question 10 [2]
Consider the circuit diagram below.



10	Determine the equivalent resistance R_{eq}. [2]

$\Downarrow$



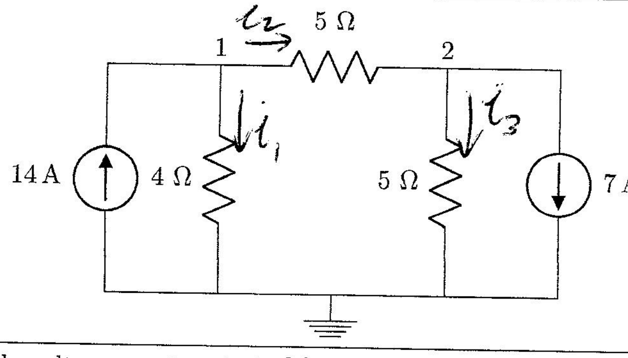
$$R_Y = R_\Delta/3$$
$$= 6/3 = 2\,\Omega$$

$$R_x = (2+6)//(2+6) = \frac{8\times 8}{8+8} = \frac{64}{16} = 4\,\Omega$$

$$R_{eq} = 3 + 2 + 4 = 9\,\Omega$$

Questions 11 to 12 [2]

Consider the circuit below.



11	Solve for the node voltage v_1 at node 1. [1]
12	Solve for the node voltage v_2 at node 2. [1]

KCL node 1

$$i_1 + i_2 = 14$$

$$\frac{v_1}{4} + \frac{v_1 - v_2}{5} = 14$$

$$(\times 20) \quad 5v_1 + 4v_1 - 4v_2 = 280$$

$$9v_1 - 4v_2 = 280 \qquad (1)$$

KCL node 2

$$i_2 = i_3 + 7 \quad \therefore \frac{v_1 - v_2}{5} = \frac{v_2}{5} + 7$$

$$(\times 5) \quad v_1 - v_2 = v_2 + 35$$

$$v_1 - 2v_2 = 35$$

$$v_1 = 2v_2 + 35 \qquad (2)$$

(2) in (1)

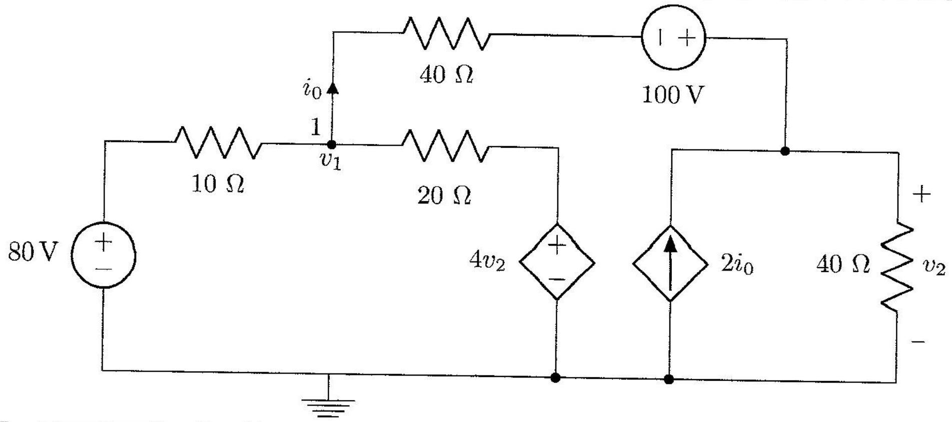
$$9(2v_2 + 35) - 4v_2 = 280$$

$$18v_2 + 315 - 4v_2 = 280 \quad \therefore 14v_2 = -35 \quad v_2 = -2.5\,V$$

$$v_1 = 30\,V$$

Questions 13 to 14 [2]

Consider the circuit below.



Use nodal analysis to set up an equation for node 1 of the form

$$av_1 - 9v_2 = b.$$

Calculate the unknown coefficients.

13	Determine a. [1]
14	Determine b. [1]

Node 1

$$i_0 + i_1 + i_2 = 0$$

$$\frac{v_1 - (v_2 - 100)}{40} + \frac{v_1 - 80}{10} + \frac{v_1 - 4v_2}{20}$$

$$(\times 40) \quad v_1 - v_2 + 100 + 4v_1 - 320 + 2v_1 - 8v_2 = 0$$

$$7v_1 - 9v_2 = +220 \quad \therefore \quad a = 7, \; b = +220$$

This page is intentionally left open for rough work.

Total Section 1: [15]

Grand Total : [15]

PRACTICE ANSWER SHEET

Assessment ID	2	0	1	9	E	B	N	1	2	2	C	0	1

	Answer	Unit
01	$P_r =$ −60	W
02	1 or 2 2	No units
03	$E_L =$ 0.04	kWh
04	$l =$ 4	No units
05	$v_x =$ 10	V
06	$v_y =$ 12	V
07	$i_x =$ 0.6	A
08	$i_1 =$ 36	A
09	$v_2 =$ 2400	V
10	$R_{eq} =$ 9	Ω
11	$v_1 =$ 30	V
12	$v_2 =$ −2.5	V
13	$a =$ 7	No units
14	$b =$ 220	No units